

Required Tasks in Image Processing

1. Point operation :

- 1- Addition
- 2- Subtraction
- 3- Division
- 4- Complement

2. Color image operation :

- 5- Changing the image lighting color
 - a) Change red
- 6- Swapping image channels:
 - a) Swap from R to G.
- 7- Eliminating color channels:
 - a) Eliminate Red

3. Image histogram :

- 8- Histogram Stretching
 - a) gray image.
- 9- Histogram Equalization.
 - a) gray image.

4. Neighborhood processing:

- 1- Linear filter :-
 - a. Average filter
 - b. Laplacian filter
- 2- Non linear filters :
 - a. Maximum;
 - b. Minimum.
 - c. Median.
 - d. Most frequent values (mode).

5. Image Restoration:

- 1- Salt and pepper noise.
 - i. Average filter
 - ii. Median filter .
 - iii. An outlier method.
- 2- Gaussian noise
 - i. Image averaging
 - ii. Average filter

6. Image segmentation:

- 1- Thresholding
 - a) Basic Global Thresholding
 - b) Automatic thresholding
 - c) Adaptive thresholding

7. Edge Detection

- 2- Sobel detector

8. Mathematical Morphology

- 3- Image dilation
- 4- Image erosion
- 5- Image opening
- 6- Boundary Extraction:
 - a) Internal boundary
 - b) External boundary
 - c) Morphological gradient

* All tasks must be collected in one package with GUI .

* For each task Display input and result images in subplot .

Name	Department	Group Mark	Individual Mark
1.			
2.			
3.			
4.			